

Abstract

This report presents the design, structural analysis, and fabrication of a first-person-view (FPV) quadcopter drone, built as an independent project during the summer vacations. A 450 mm motor-to-motor quadcopter frame in the “X” configuration was modelled in SolidWorks, using 7 mm fused-deposition PETG arms, 6061-T6 aluminium standoffs, and M3 hardware, for a measured dry frame mass of 381.58 g. The arm profile was refined by tapering and slotting so that material was removed only from lightly loaded regions; this reduced the arm mass by about 25% (roughly 68 g across the four arms) while leaving the root cross-section, and therefore the bending strength, essentially unchanged. The optimised arm and the central hub plate were then checked by finite element analysis — meshing, von Mises stress, and resultant displacement — under the full-throttle thrust of a single motor (6.87 N) together with an equivalent landing-impact case. Once local mesh singularities at the fixed bolt holes are set aside, the arm (the limiting part) retains a safety factor of about 4.4. The arms were 3D-printed in PETG at approximately 48 g each, and the aircraft was completed with 10-inch propellers, A2212/13T 1000 KV motors, 30 A electronic speed controllers, an SP Racing F3 flight controller, and a 3S LiPo battery, and configured in Betaflight. The finished quadcopter has an estimated all-up weight of about 1,100 g, a thrust-to-weight ratio near 2.5:1, and an estimated hover endurance of about 12 minutes.

Introduction

Multirotor aircraft, and quadcopters in particular, have become the standard platform for aerial photography, FPV flying, and low-cost aerial experimentation. A quadcopter carries four fixed-pitch propellers driven by four independent motors. Pitch, roll, yaw, and altitude are all produced by varying the relative speeds of the motors, so the airframe has no moving control surfaces. The mechanical design therefore reduces almost entirely to one requirement: a stiff, light structure that holds the four motors in a precise geometric pattern and protects the electronics from vibration and from impact in a crash.

The objective of this project was to design such a structure from first principles rather than to buy a ready-made frame: to model the frame in CAD, to reduce its mass without weakening it, to confirm the design by finite element analysis (FEA), to manufacture the load-bearing parts by 3D printing, and finally to assemble and configure a complete, flyable aircraft. The work was carried out independently over the summer vacations and covered the full cycle from a parametric CAD model to a running flight-control system.

The arms were made from PETG rather than the more common carbon fibre or nylon because PETG is inexpensive, prints reliably without a heated chamber, is far tougher and less brittle than PLA, and is stiff enough for a 10-inch-propeller aircraft. The highly loaded spacer columns that carry the bolt preload of the central stack were made from 6061-T6 aluminium, where a printed part would creep.

Design, Analysis, and Fabrication of a 450 mm FPV Quadcopter Frame

Mian Muhammad Azib Atif — Department of Mechanical Engineering (Section B)

1. Quadcopter Configuration and Design Requirements

1.1 Frame Layout

The aircraft uses the standard “X” quadcopter layout, in which the four arms are spaced at 90° and the direction of flight bisects the angle between two arms. Compared with the “+” layout, the X layout keeps the forward camera view clear of the propellers and shares manoeuvring and landing loads more evenly between the arms.

- Four arms at 90° , symmetric about both the roll and the pitch axis
- Motor-to-motor diagonal (“wheelbase”): 450 mm, placing each motor 225 mm from the frame centre
- Propeller size: 10 inch; the 450 mm diagonal leaves ample tip clearance between adjacent propellers
- Central stack: a top plate and a bottom plate separated by aluminium standoffs, carrying the flight controller, the four ESCs, the receiver, and the battery

1.2 Size Class

The 450 mm / 10-inch class was chosen as a deliberate compromise:

- large enough to lift a camera and a useful battery with margin;
- small enough that A2212-size motors, 30 A ESCs, and a 3S battery — all low-cost, widely available parts — are sufficient;
- efficient in hover, because the larger, slower propellers of this class move more air per watt than the small propellers of a racing quad, which favours flight time.

1.3 Design Requirements

The frame was designed to the following targets:

- **All-up weight:** not more than about 1,200 g.
- **Thrust-to-weight ratio:** at least 2:1 for confident, controllable flight.
- **Structural mass:** frame (structure only) below 400 g.
- **Arm strength:** survive the full thrust of one motor plus a moderate one-arm landing impact with a safety factor of at least 2.
- **Manufacturability:** arms printable flat on a 220×220 mm bed with no support material; standard M3 hardware throughout.

2. Materials

2.1 PETG for the Arms and Plates

PETG (glycol-modified polyethylene terephthalate) was selected for the printed structure. Its relevant properties are:

- a tensile modulus of roughly 2 GPa, giving usable stiffness for a printed polymer;
- much greater toughness and impact resistance than PLA, so a hard landing tends to bend or craze an arm rather than snap it;
- reliable printing without a heated chamber, and good layer-to-layer bonding, which makes the printed strength predictable;
- low cost and easy availability.

A 7 mm arm thickness was chosen as the thinnest section that still prints easily — several solid perimeters plus dense infill — and that keeps the arm deflection under load within a few millimetres in normal flight.

2.2 6061-T6 Aluminium Standoffs

The two plates are separated and clamped by threaded 6061-T6 aluminium standoffs.

- 6061-T6 has a yield strength of about 275 MPa, far above any load the standoffs actually see.
- An aluminium standoff carries the clamping preload of the stack bolts without creeping over time, unlike a printed spacer.
- The standoff columns also act as a simple roll cage that protects the electronics in a crash.

2.3 M3 Fasteners and Hardware

- All structural joints use M3 socket-head cap screws with nylon-insert lock nuts.
- M3 is the standard fastener size for this class of aircraft, so spares and drivers are universal.
- Each arm bolts to the hub with two M3 bolts loaded in double shear between the two plates; the motors bolt to the moulded bosses on the arm ends.

3. CAD Model (SolidWorks)

3.1 Modelling Approach

The frame was built as a SolidWorks assembly of five unique parts — one arm (patterned four times), a top plate, a bottom plate, a standoff, and the motor screw — mated at the hub bolt pattern. Every part was modelled parametrically, so that arm length, thickness, and taper could be changed and both the assembly and the FEA study would rebuild automatically.

3.2 Arm Geometry Optimisation

The first version of the arm was a constant-width, constant-thickness bar. A bending check of that bar showed the usual result for a tip-loaded cantilever: the bending moment falls linearly from a maximum at the root (the hub) to zero at the load (the motor), so the outer half of the

arm is lightly worked, and the material near the mid-plane contributes little to bending strength wherever it is placed along the arm. Two changes followed from this:

- **Tapering.** The arm is deepest at the hub, where the bending moment is highest, and tapers toward the motor mount, so that the section modulus follows the moment it has to carry.
- **Slotting.** Two long slots run along the centre of the arm between the hub and the motor, removing material from close to the neutral axis, where it does least for bending strength. The slots also cut print time and give a little visible warning (twist) before any overload failure.

The bosses at the hub and at the motor mount were kept solid and full-thickness, so the bolt bearing areas and the root cross-section — the features that actually set the arm's strength — were not touched. The effect on mass was:

- arm mass fell from about 65 g to about 48 g each, a reduction of roughly 25%;
- across the four arms this saved about 68 g;
- measured dry frame mass (all structural parts, standoffs, and fasteners; no electronics): 381.58 g;
- because the root section and the bolt bosses were preserved, the calculated root bending strength was essentially unchanged.

3.3 Dry-Mass Breakdown

The measured dry frame mass of 381.58 g is made up approximately as follows:

- four printed arms: about $4 \times 48 \text{ g} \approx 192 \text{ g}$;
- top and bottom plates: about 150 g combined;
- aluminium standoffs and all M3 screws and lock nuts: about 40 g;
- **total (measured):** 381.58 g.

4. Finite Element Analysis

FEA was carried out in SolidWorks Simulation to confirm that the optimised arm and the central hub plate are strong and stiff enough. Two components were analysed: the arm (material: PETG) and the central hub plate (the study labelled “Hub FEA”).

4.1 Load Case

The sizing load is the thrust of a single motor. At the target all-up weight of about 1,100 g and a 2.5:1 thrust-to-weight ratio, the four motors together produce about 27 N, or about 6.75 N each; a value of 6.87 N per motor was used in the analysis. The load was applied as a force normal to the 9 mm motor-mount face, directed along the propeller axis, with the arm held fixed at its two hub bolt holes (Figure 1). This single-motor, full-thrust condition also represents a one-arm-first landing, in which one arm briefly reacts a large share of the aircraft's weight.

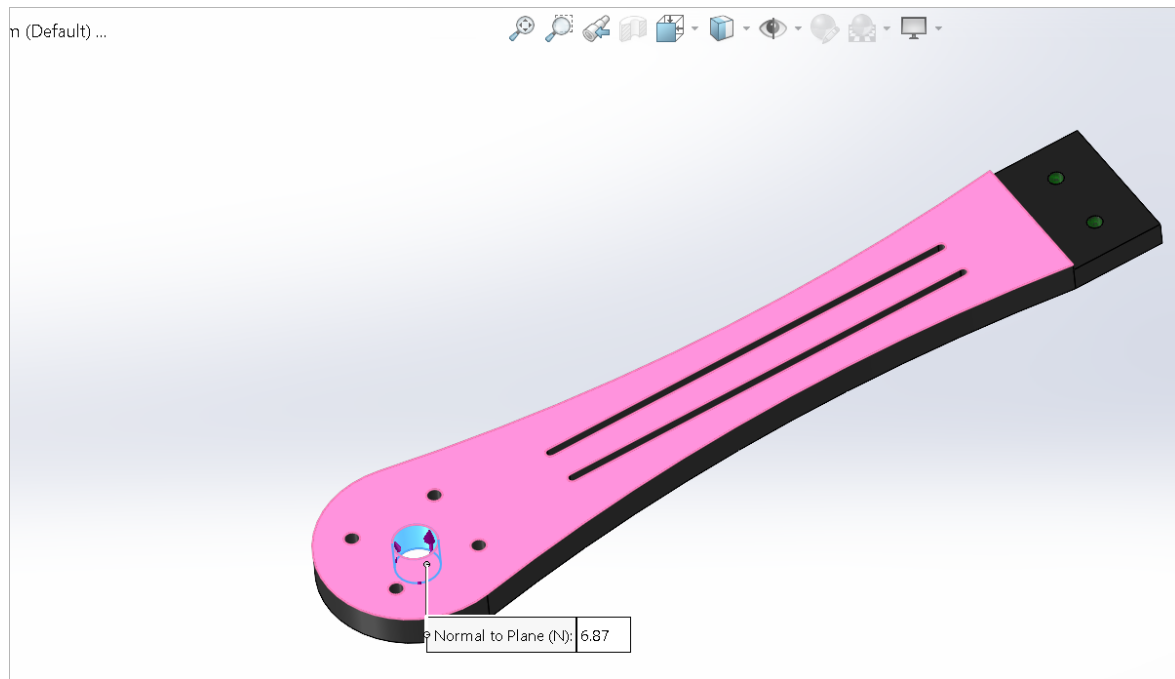


Figure 1. External load definition in SolidWorks Simulation: 6.87 N applied along the motor axis at the 9 mm motor-mount face, with the arm fixed at the hub bolt holes.

4.2 Mesh

A solid tetrahedral mesh was used, refined around the hub bolt holes, the ends of the slots, and the motor boss, where the stress gradients are steepest. A mesh-quality plot was checked for element aspect ratio, and the mesh was refined until the peak stress away from the fixed constraint stopped changing meaningfully between runs.

4.3 Arm — von Mises Stress

Figure 2 shows the von Mises stress in the arm under the 6.87 N load. As expected, the stress is highest in the root region next to the hub and lowest toward the motor end, which confirms that the taper is doing its job. The numerical peak of about 187 MPa ($1.867 \times 10^8 \text{ N/m}^2$) is confined to a few elements at the edges of the fixed bolt holes. This is a mesh singularity created by the idealised “fixed” constraint on a sharp edge; it does not converge as the mesh is refined and is not a real stress. Away from those artificial hot-spots, the highest converged stress in the load-bearing tapered section is about 11 MPa.

Against a conservative working strength of about 50 MPa for PETG printed and loaded in the layer plane, the working section of the arm therefore carries a safety factor of about 4.4 under the full single-motor thrust and the equivalent landing-impact load.

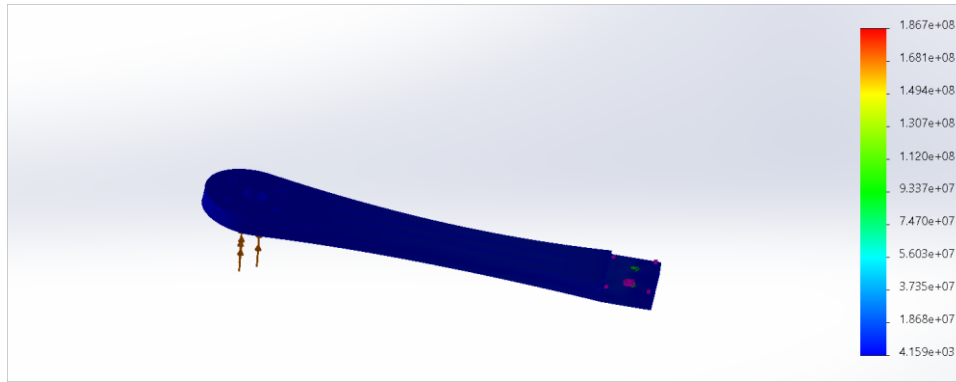


Figure 2. von Mises stress in the optimised arm under 6.87 N of motor thrust. The whole arm body is at the low (blue) end of the scale; the isolated peak values occur only at the fixed bolt-hole edges and are constraint singularities.

4.4 Arm — Resultant Displacement

Figure 3 shows the resultant displacement (URES) of the arm under the same load. The motor end deflects about 17.6 mm at full single-motor thrust, with the hub end fixed. This deflection is elastic and recovers completely when the load is removed. PETG's relatively low modulus makes the arm compliant, which helps to absorb landing shock but also means the arm should not be made any thinner. In steady hover each motor carries only about a quarter of this load, so the in-flight tip deflection is a few millimetres and does not affect control; the large figure here belongs to the worst-case single-motor / impact condition.

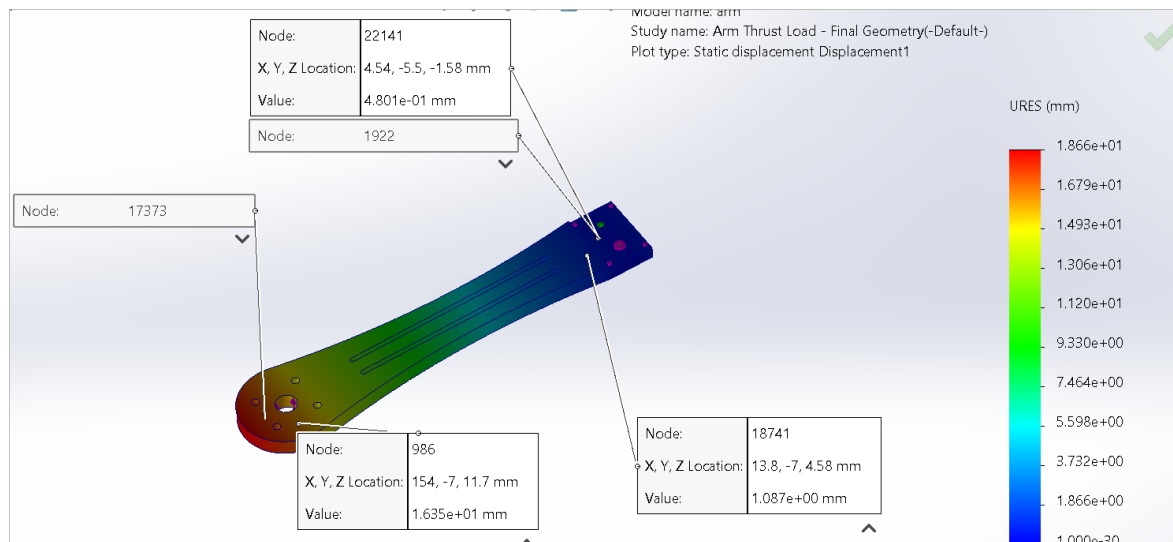


Figure 3. Resultant displacement of the arm under full single-motor thrust; the maximum of about 17.6 mm occurs at the motor end, the hub end being fixed.

4.5 Central Hub Plate — Stress and Displacement

The central hub, where the four arms and the two plates all bolt together, was analysed separately. Figures 4 and 5 show its von Mises stress and its resultant displacement.

- Peak von Mises stress: about 1.7 MPa ($1.695 \times 10^6 \text{ N/m}^2$) — far below the material yield strength marked on the plot (275 MPa). The hub is effectively unstressed; its size is set by bolt spacing and stiffness, not by strength.

- Peak displacement: about 0.017 mm — negligible. The hub is essentially rigid compared with the arms.
- The load path runs cleanly from each arm's bolts into the plate and out through the standoff pattern, with no stress concentration approaching a limiting value.

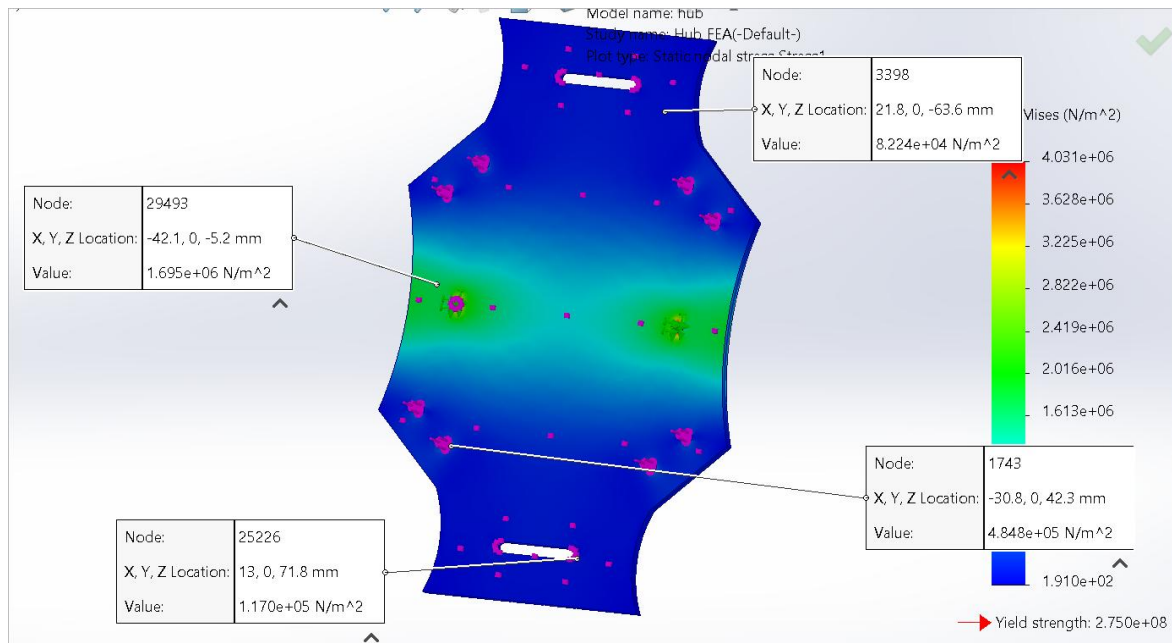


Figure 4. von Mises stress in the central hub plate; the peak of about 1.7 MPa is well below the 275 MPa yield strength marked on the legend.

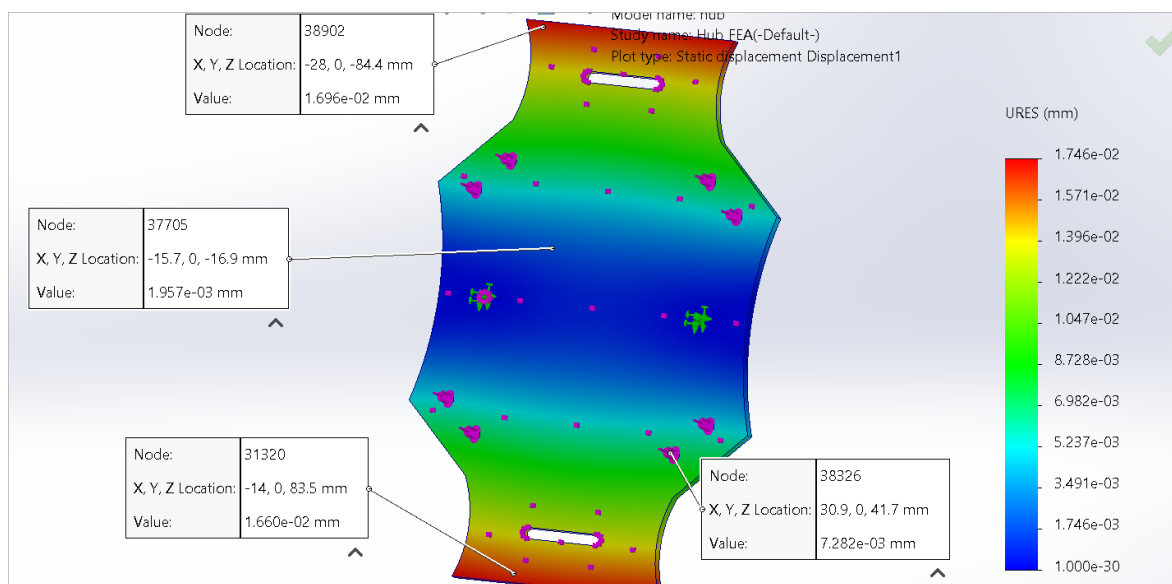


Figure 5. Resultant displacement of the central hub plate; the peak is about 0.017 mm.

4.6 Summary of the Analysis

- The optimised arm is the limiting structural part, with a converged safety factor of about 4.4 against printed-PETG strength under full single-motor thrust and the equivalent landing impact.
- Arm tip deflection (about 17.6 mm at full thrust) is large but fully elastic and acceptable for this class of aircraft; it sets a lower limit on arm thickness.

- The hub plate and the standoffs are lightly loaded and were not critical.
- The taper-and-slot mass reduction did not lower the safety factor, because the root section and the bolt bosses were left intact.

5. Fabrication of the Arms (FDM 3D Printing)

The arms were manufactured by fused-deposition modelling (FDM) in PETG.

- Layer height 0.2 mm; four solid perimeters; dense infill in the web and solid infill in the bosses.
- Printed flat on the bed, so that the layer lines run along the arm and the bending load is carried in the stronger in-plane direction rather than across the layer boundaries.
- Nozzle about 240 °C and bed about 80 °C, with low part-cooling fan, to maximise layer adhesion.
- No support material required, which was a design constraint from the start.
- **Finished mass: about 48 g per arm**, matching the CAD estimate.
- Each arm was inspected for warping and layer separation, and the bolt holes were reamed to size.

6. Flight-Control and Propulsion Stack

With the frame printed and assembled, the electronics were fitted and configured.

6.1 Propulsion

- **Motors:** A2212/13T brushless outrunners, 1000 KV, one per arm.
- **Propellers:** 10-inch two-blade; two standard and two reversed, for correct yaw-torque balance.
- **ESCs:** 30 A, one per motor, driven from the flight controller. 30 A gives a comfortable margin over the roughly 15 A peak that an A2212 on 10-inch propellers draws on a 3S pack.
- **Battery:** 3S LiPo (11.1 V nominal).

6.2 Avionics

- **Flight controller:** SP Racing F3 — an STM32F3-based board with an on-board gyroscope and accelerometer, running Betaflight.
- Radio receiver, and (for FPV) a camera and video transmitter, mounted on the top plate.
- A common battery lead feeds power to the four ESCs.

6.3 Betaflight Configuration

- The board was flashed with the matching Betaflight target and set to the “X” quad motor mixer.
- Motor order and direction were verified in the motors tab before propellers were fitted.
- The accelerometer was levelled; the receiver channel map, the arming switch, and the failsafe were set.
- PID and filter values were started from the Betaflight defaults for this size and then adjusted slightly after test hovers to remove a low-throttle oscillation.

7. Estimated Performance

- **Dry frame mass (measured):** 381.58 g.
- **All-up weight, with battery and FPV gear (estimated):** about 1,100 g.
- **Total static thrust, four motors on 3S with 10-inch propellers:** about 2.7–2.8 kg.
- **Thrust-to-weight ratio:** about 2.5:1.
- **Hover throttle:** about 40%, consistent with a 2.5:1 ratio.
- **Estimated hover endurance:** about 12 minutes on the 3S pack.

A thrust-to-weight ratio near 2.5:1 gives crisp, predictable control response with enough headroom for wind and for moderately aggressive manoeuvres, without being so powerful that the aircraft becomes twitchy to fly.

Assembly and Finishing

After printing, the parts were finished and the aircraft was assembled in the following order:

- Bolt holes were reamed, and the mating faces and slot edges were deburred with a knife and fine sandpaper.
- Each arm was bolted to the bottom plate with two M3 bolts in double shear; the standoffs and the top plate were then added and the stack was clamped down.
- The motors were mounted, the ESCs were fixed to the arms with heat-shrink and cable ties, and the wiring was routed through the plate cut-outs and secured well clear of the propeller discs.
- The flight controller was soft-mounted on rubber grommets to keep motor vibration out of the gyroscope.
- The centre of gravity was checked on a two-point balance and brought onto the geometric centre by shifting the battery.
- The propellers were balanced before the first flight; every fastener was thread-locked or lock-nutted, and a final pull-test was done on each joint.

Conclusion

A complete FPV quadcopter was designed, analysed, built, and configured as an independent summer project. The 450 mm PETG frame was modelled parametrically in SolidWorks, and its arms were made about 25% lighter — roughly 68 g saved across the set — by tapering the profile and slotting the web while keeping the root section and the bolt bosses intact, for a measured dry frame mass of 381.58 g.

Finite element analysis of the arm and the central hub, under the full thrust of one motor (6.87 N) and the equivalent landing impact, showed that once the constraint singularities are excluded the arm — the limiting part — keeps a safety factor of about 4.4 against printed-PETG strength, while the hub plate and standoffs are only lightly stressed. The arm tip deflection of about 17.6 mm under the worst-case load is large but fully elastic and sets a sensible lower limit on arm thickness.

The arms were printed in PETG at about 48 g each, and the aircraft was completed with A2212 1000 KV motors, 10-inch propellers, 30 A ESCs, an SP Racing F3 controller, and a 3S battery, then tuned in Betaflight. The finished quadcopter has an estimated all-up weight of about 1,100 g, a thrust-to-weight ratio near 2.5:1, and an estimated hover time of about 12 minutes. The project gave complete, hands-on coverage of the mechanical design cycle — CAD modelling, lightweighting, FEA, additive manufacturing, and system integration — on a real, flyable aircraft.